JAGMEET SINGH

AI Engineer | Full-Stack & LLM Systems
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SUMMARY

AI Engineer at 1GEN building full-stack platforms and tool-using LLM agents. I own features from database schema and API design through web UI, CI/CD and AWS deployment. Recent work: a multi-tenant school SaaS (NestJS, PostgreSQL, Next.js) with Square payments, seat-based licensing, SSO and analytics; an AI HR agent with zero-knowledge salary encryption; Claude-based chat, WhatsApp and voice agents; and a self-healing incident agent (with its own PyPI SDK) that opens GitHub fix PRs.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (Node.js), SQL, Go (Gin)

Backend: NestJS, Express, Flask, REST APIs, WebSockets, Prisma ORM, cron and queue workers, PDF generation

Frontend: Next.js, React, Tailwind CSS, GSAP; Kotlin Compose Multiplatform (project exposure)

AI / LLM: Anthropic Claude and Google Gemini tool/function calling, agent orchestration, LLM fallback design, Gemini embeddings, ElevenLabs STT/TTS, Amazon Polly

Databases: PostgreSQL, MongoDB, Redis, MySQL

Cloud & DevOps: Docker, Docker Compose, GitHub Actions, GHCR, Nginx, AWS (EC2, RDS, S3, STS, SES, SNS, Polly)

Integrations: Square payments and webhooks, Twilio (SMS, WhatsApp, Voice/IVR), Slack API, Google OAuth, Clever SSO, Firebase FCM, GitHub REST API, Google Sheets

Security & Testing: JWT refresh-token rotation, OAuth ID-token verification, RSA-OAEP + AES-GCM hybrid encryption, webhook HMAC verification; Jest, Vitest, pytest

Data & Tools: Pandas, NumPy, Power BI, Selenium, Git, AI-assisted development (Claude Code)

EXPERIENCE

AI Engineer, 1GEN — Jan 2025 – Present

Joined as Data Scientist. Delivered across 8 products, working on backend, frontend, AI agents and deployment.

Mind Circuit / Ease – social-emotional wellness platform for schools (in production) — Jul 2026 – Present

Next.js, React, TypeScript, NestJS, Prisma, PostgreSQL, Redis, AWS (EC2, RDS, S3, SES, Polly), Docker, GitHub Actions | 5 repositories

- **Lead engineer across 5 repositories:** teacher web portal, NestJS API, ops console, MCAP analytics app and school-admin backend. Built the teacher portal as a new web client of the mobile API (auth, onboarding, activity player, built-in reset activities, classroom analytics).
- **Payments & licensing:** built the Square layer end to end, covering SCA/3DS card checkout, HMAC-verified webhooks, idempotent payment records reconciled by a background resolver, and renewals claimed with SELECT … FOR UPDATE SKIP LOCKED. Also built 14-day trials, seat-based school licences, entitlements and PDF invoices.
- **Multi-tenant school administration:** tenancy guard, admin approval queue, school-scoped analytics, and a Postgres advisory lock that stops duplicate scheduled emails.
- **Auth & security:** added Google ID-token verification to stop duplicate OAuth accounts, provider linking that refuses unverified email matches, and Clever SSO role mapping. Found and fixed an API endpoint that returned users' bcrypt password hashes.
- **MCAP analytics:** built engagement, implementation, and payments & licensing dashboards and their APIs. Moved MCAP out of the admin console into a standalone app with server-enforced access control.
- **Release pipeline:** replaced building on the server with GitHub Actions: a test gate, then GHCR images pinned per commit, auto-migrations on deploy, and production Compose for 4 apps on EC2 + RDS over TLS. Profiled the build and cut Next.js peak build memory by 61% (2.30 GB → 0.89 GB). Wrote ~30 Jest spec suites covering payments, licensing and auth.

Geny / OneOS – AI HR assistant for web chat and Slack (in internal production) — Feb 2026 – Jun 2026

Python, Flask, Google Gemini, MongoDB, Slack API, AWS (SES, S3), pytest

- **Owned the backend and agent layer** of an HR assistant. Employees use it to apply for leave, raise tickets and manage teams, salary structures, HR letters and inventory.
- **Agent design:** built an LLM function-calling intent engine on a declarative capability registry. Each of 14 HR actions is defined once (schema, API mapping, auto-filled fields, confirmation step), so new actions ship without code changes. Migrated the LLM backend from Claude to Gemini.
- **Zero-knowledge salary data:** admins hold non-extractable RSA-2048 keys in the browser. Each response is hybrid-encrypted with a fresh AES-256-GCM key, with per-admin isolation and approved key rotation, so the server never holds a decryption key.
- **Integrations:** Slack bot with manager approve/reject actions inside Slack's 3-second window, bulk salary-slip PDF generation and email via AWS SES, S3 storage and Google Sheets sync. Wrote 130+ pytest tests.

Wingz – private-aviation charter platform Apr 2026 – Sep 2026

Node.js, Express, Next.js, PostgreSQL, Anthropic Claude, Twilio, ElevenLabs, WebSockets; Kotlin Compose Multiplatform

- **AI agents:** built a Claude tool-use website concierge with separate anonymous and signed-in toolsets (availability, quotes, leads, tickets). Built a WhatsApp booking agent behind a provider adapter that switches between Twilio and Meta.
- **LLM-outage fallback:** errors are classified into customer-safe messages, a deterministic booking flow still captures the lead, and failure reasons are shown to staff in the CRM.
- **Jarvis voice assistant:** WebSocket proxy that streams ElevenLabs real-time speech-to-text, plus low-latency streaming TTS, barge-in and audit logging. API keys stay server-side.
- **Internal tools:** built a CRM (leads, inbox, tickets, bookings, analytics) and a revenue tool for quotes, GST proforma/tax invoices, receipts, operator bills and payouts, with PDF generation.
- **Security & mobile:** built JWT refresh-token rotation (token families, reuse detection, row-locked rotation; 15 tests); fixed an IDOR and a DB pool-poisoning bug. Added auth and empty-leg seat bidding to the Kotlin Multiplatform app.

AIOps Self-Healing Agent – automated incident detection and fix proposals Feb 2026 – Mar 2026

Python, Flask, Anthropic Claude / Gemini, MongoDB, Slack API, GitHub REST API, PyPI packaging

- **Published aiops-sdk on PyPI (7 releases):** a Python client with Flask integration that captures unhandled exceptions, 4xx/5xx responses and ERROR logs, and sends heartbeats every 30 seconds.
- **Built v1 solo:** exception ingestion, rule-based P0–P3 severity, SHA-256 fingerprint deduplication, and an orchestrator driving auditor, fixer and reporter agents, with Slack approve/reject before any change.
- **Fix pipeline:** built a client-side code probe that holds no LLM or secrets. It locates the failing route from the stack trace, returns sanitized code context, and runs candidate fixes against the service's tests in a sandbox with rollback. A GitHub service then opens the fix PR; it produced working auto-generated PRs in a live demo.

Luminous Super App (prototype) Apr 2026

Go, Gin, PostgreSQL, Redis, Kafka, Nginx, React, Twilio

- Added a scale and security layer to Go microservices: rate limiting, refresh tokens, Redis Pub/Sub chat fan-out, wallet fraud checks, a Kafka event producer, Postgres full-text search and PgBouncer. Also added Twilio SMS OTP auth and a rider-delivery mini-app.

GC Agent – AI general-contractor platform (demo) May 2026

Python, Flask, Anthropic Claude, Twilio Voice, MongoDB

- Took over the prototype on a demo deadline and made it demo-ready: Twilio signature verification, idempotent callbacks, DTMF IVR, a stateful LLM chat intake, background workers, and 20+ critical fixes verified with end-to-end scripts.

Nano Pe – fintech lending app (technical lead) Dec 2025

Python, Flask, MongoDB, AWS SNS, JWT

- Led a small backend team (architecture, code reviews, deployment) that delivered loan eligibility, offers based on a custom credit score (N-Score), EMI tracking, and penalty and grace-period handling.
- Built phone-OTP signup and login (AWS SNS), JWT sessions, profile completion and loan initiation APIs myself.

1GEN Impact Cloud – product suite for NGOs and organizations (in production) May 2025 – Dec 2025

Python, Flask, MongoDB, Google Gemini, Firebase, AWS SNS

- Built backend APIs across the suite: campaign manager with AWS SNS SMS and email delivery stats, pitch builder, sprint dashboards, ticket analytics (resolution time, assignee performance), crowdfunding, careers and leave analytics. Built Reciprocity matching on Gemini embeddings.
- **Silver Talkies (community app for seniors):** event, meetup and workshop booking, travel booking, feeds and stories, Firebase push notifications, and encrypted document handling.
- **TalentHub:** delivered a medical-coding learning and assessment platform in one week (OTP login, courses, assessments, progress tracking).

Data Analyst Intern (Part-time), Candiclie Consultancy Sep 2024 – Nov 2024

- Collected and validated data from multiple sources, and built Selenium WebDriver scrapers for analysis datasets.

Data Analyst Intern, Mentorness Jun 2024 – Jul 2024

- Analyzed hotel-reservation data in SQL for booking and guest-preference trends, built Power BI dashboards on YouTube song data, and presented the findings to stakeholders.

Fundraising Intern, Free Pads for India (NGO) Jul 2024 – Aug 2024

- Planned crowdfunding campaigns and brought in new donors and partners for a menstrual-health awareness initiative.

EDUCATION

B.Tech, Computer Science & Engineering – Quest Group of Institutions, Jhanjeri, Punjab 2025 | CGPA 8.7

Diploma, Computer Science & Engineering – Govt. Polytechnic College, Khunimajra, Punjab 2022 | CGPA 8.2

CERTIFICATIONS

- BCG GenAI Job Simulation (Forage) | Career Essentials in Generative AI (Microsoft & LinkedIn)

Languages: English, Hindi, Punjabi